

---

# **COSMETIC INSIGHTS:** NAVIGATING COSMETICS TRENDS AND CONSUMER INSIGHTS WITH TABLEAU

---

## **PROJECT TEAM MEMBERS**

**Bandila Jyothi** – Team Lead

**Pasupuleti TagorePremReddy** – Member

**Bharath Dadala** – Member

**Varshini Siddireddy** – Member

**Sai Durga Amrutha Vakada** – Member

| S.No     | Contents                                  |
|----------|-------------------------------------------|
| <b>1</b> | <b>INTRODUCTION</b>                       |
| 1.1      | Project Overview                          |
| 1.2      | Purpose                                   |
| <b>2</b> | <b>IDEATION PHASE</b>                     |
| 2.1      | Problem Statement                         |
| 2.2      | Empathy Map Canvas                        |
| 2.3      | Brainstorming                             |
| <b>3</b> | <b>REQUIREMENT ANALYSIS</b>               |
| 3.1      | Solution Requirement                      |
| 3.2      | Data Flow Diagram                         |
| 3.3      | Technology Stack                          |
| <b>4</b> | <b>PROJECT DESIGN</b>                     |
| 4.1      | Proposed Solution                         |
| 4.2      | Solution Architecture                     |
| <b>5</b> | <b>PROJECT PLANNING &amp; SCHEDULING</b>  |
| <b>6</b> | <b>FUNCTIONAL AND PERFORMANCE TESTING</b> |
| <b>7</b> | <b>RESULTS</b>                            |
| 7.1      | Chart Visualizations                      |
| 7.2      | Dashboard Screenshots                     |
| 7.3      | Story Screenshot                          |
| 7.4      | Tableau Public Profile                    |

| <b>S.No</b> | <b>Contents</b>                       |
|-------------|---------------------------------------|
| <b>8</b>    | <b>ADVANTAGES &amp; DISADVANTAGES</b> |
| <b>9</b>    | <b>CONCLUSION</b>                     |
| <b>10</b>   | <b>FUTURE SCOPE</b>                   |
| <b>11</b>   | <b>APPENDIX</b>                       |

# 1. INTRODUCTION

The cosmetics industry has experienced significant growth due to increasing consumer awareness, changing beauty trends, and the availability of a wide range of cosmetic products. Customers often face difficulties in selecting suitable products because of the large number of brands, product categories, price variations, and skin-type requirements. Analyzing such large amounts of cosmetic data manually is time-consuming and inefficient.

The project "Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau" was developed to analyze cosmetic products and consumer preferences using data visualization techniques. The project utilizes Microsoft Excel for data preparation and Tableau Public for creating interactive dashboards and stories. These visualizations help users understand product performance, brand trends, pricing strategies, skin-type analysis, and customer insights.

The project aims to transform raw cosmetic data into meaningful information through interactive charts, dashboards, and stories. This enables customers and businesses to make better decisions based on data-driven insights. The use of Tableau provides an effective and user-friendly approach to analyzing cosmetic market trends and consumer behavior.

## 1.1 Project Overview

The Cosmetic Insights project focuses on analyzing cosmetic products, brands, prices, skin types, and consumer preferences using Tableau. The project involves collecting cosmetic data, preprocessing the dataset, creating visualizations, developing dashboards, and generating stories to provide meaningful insights. Interactive dashboards help users explore market trends, compare products and brands, and understand customer preferences effectively.

This project uses **Microsoft Excel** for data collection and preprocessing, and **Tableau Public** for creating interactive visualizations, dashboards, and stories. Various analytical techniques and visual representations are used to identify market trends, compare cosmetic brands, analyze product pricing, and understand customer preferences. The dashboards provide users with an easy

and interactive way to explore cosmetic data and generate meaningful business insights.

The project helps both customers and businesses understand market behavior, identify popular brands, analyze product performance, and support data-driven decision-making through effective visualization techniques.

## **1.2 Purpose**

The primary purpose of this project is to develop an interactive analytics platform that provides valuable insights into cosmetic products, brands, and consumer preferences using Tableau. The project aims to simplify the process of analyzing cosmetic data by transforming raw datasets into meaningful visual information.

The project helps customers identify suitable cosmetic products based on brand, price, and skin type. It also enables businesses to understand market trends, customer behavior, and product performance through visual analytics. By using interactive dashboards and stories, the project supports effective decision-making and improves the understanding of cosmetic market dynamics.

Additionally, this project demonstrates the practical application of data analytics and visualization tools in solving real-world business problems and generating actionable insights for both customers and organizations.

## **2. IDEATION PHASE**

The ideation phase is one of the most important stages in project development. It involves identifying the problem, understanding customer requirements, analyzing market needs, and generating suitable solutions. During this phase, the project team studied various aspects of the cosmetics industry, including cosmetic brands, products, pricing, customer preferences, and skin-type requirements. Through analysis and discussions, the team identified the need for an interactive data visualization platform that can provide meaningful insights into cosmetic products and consumer behavior. The ideation process helped in selecting the most effective solution using Tableau dashboards and stories.

### **2.1 Problem Statement**

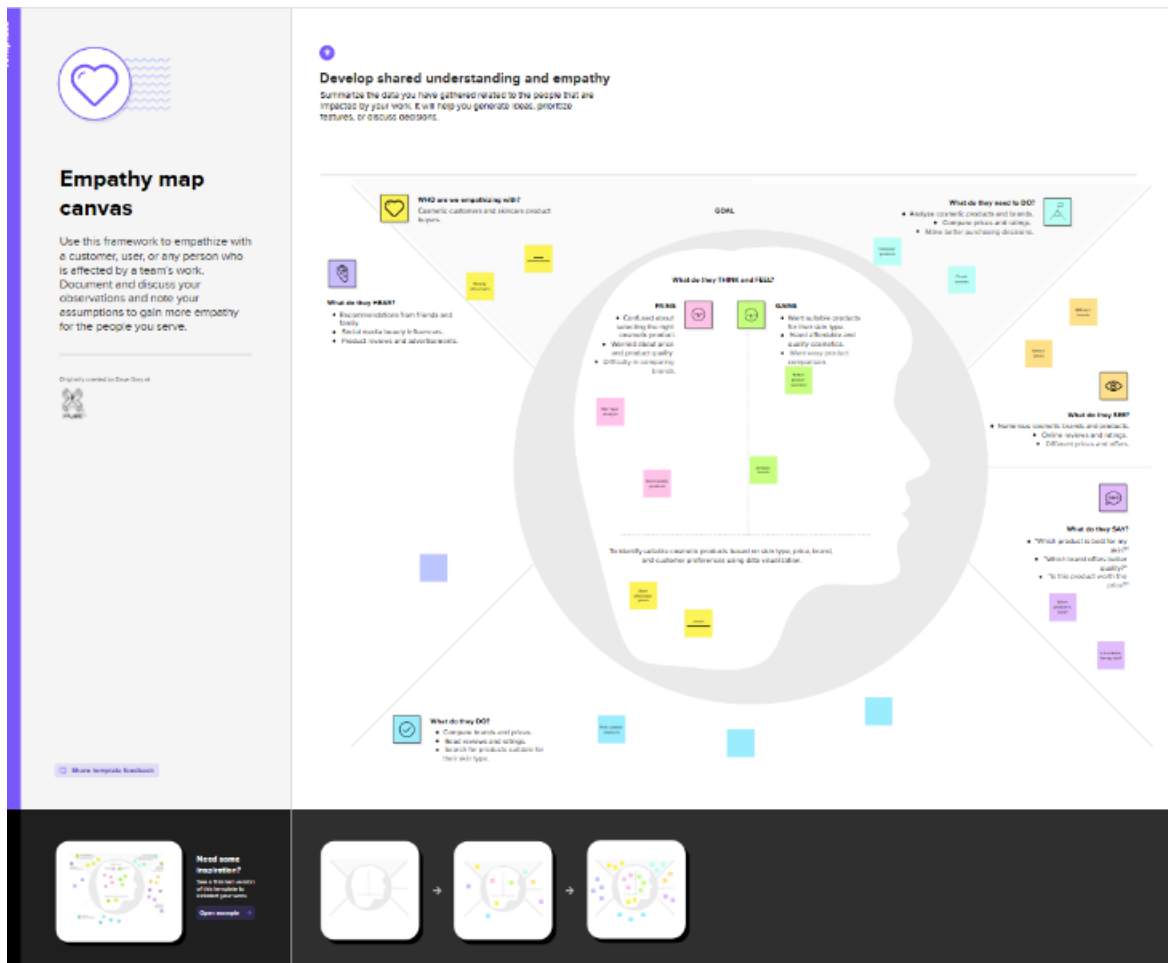
The cosmetics industry contains a wide variety of products, brands, prices, and categories, making it difficult for customers to select suitable products according to their preferences and skin types. Customers often face challenges in comparing cosmetic products, understanding pricing patterns, and analyzing market trends due to the lack of centralized information and analytical tools.

This project aims to solve these challenges by developing an interactive Tableau-based analytics system that provides insights into cosmetic products, brands, pricing strategies, skin types, and customer preferences. The project supports effective decision-making through data visualization and interactive dashboards.

### **2.2 Empathy Map Canvas**

The empathy map canvas was developed to understand customer needs, emotions, behaviors, and expectations related to cosmetic product selection. Customers often experience confusion due to the availability of numerous products and brands in the market. They require reliable information regarding product quality, pricing, brand reputation, and skin-type suitability.

The empathy mapping process helped identify customer pain points and provided valuable insights into customer behavior and preferences. This understanding assisted in designing an effective and user-friendly analytics solution using Tableau dashboards and stories.



## 2.3 Brainstorming

Brainstorming sessions were conducted to identify the key features and functionalities required for the Cosmetic Insights project. The project team discussed various ideas related to cosmetic product analysis, brand comparison, price analysis, skin-type analysis, dashboard creation, and story visualization. Different approaches were evaluated based on feasibility, usability, and project objectives.

After analyzing various ideas, the team selected Tableau as the primary visualization tool and developed interactive dashboards and stories to provide meaningful insights into cosmetic trends and consumer preferences.

### **3. REQUIREMENT ANALYSIS**

Requirement Analysis is an important phase in project development that helps identify the functional and non-functional requirements of the system. In this phase, customer requirements, system requirements, data flow, and technology components are analyzed to develop an effective solution. For the Cosmetic Insights project, the requirements were identified based on customer needs, cosmetic market analysis, and data visualization objectives. The analysis helped in designing an interactive and user-friendly analytics platform using Tableau.

#### **3.1 Solution Requirement**

The Cosmetic Insights project requires collecting and analyzing cosmetic product data to generate meaningful insights through interactive visualizations. The system should support data import, preprocessing, analysis, dashboard creation, story visualization, and publishing through Tableau Public. The project should provide accurate, reliable, and user-friendly visualizations to help customers and businesses understand cosmetic market trends and consumer behavior.

##### **Functional Requirements**

Data collection and import from Excel datasets.

Data preprocessing and validation.

Analysis of cosmetic brands, prices, categories, and skin types.

Development of interactive dashboards.

Creation of Tableau stories.

Publishing dashboards and stories using Tableau Public.

##### **Non-Functional Requirements**

User-friendly interface.

Reliable and accurate analysis.

Fast performance and dashboard loading.

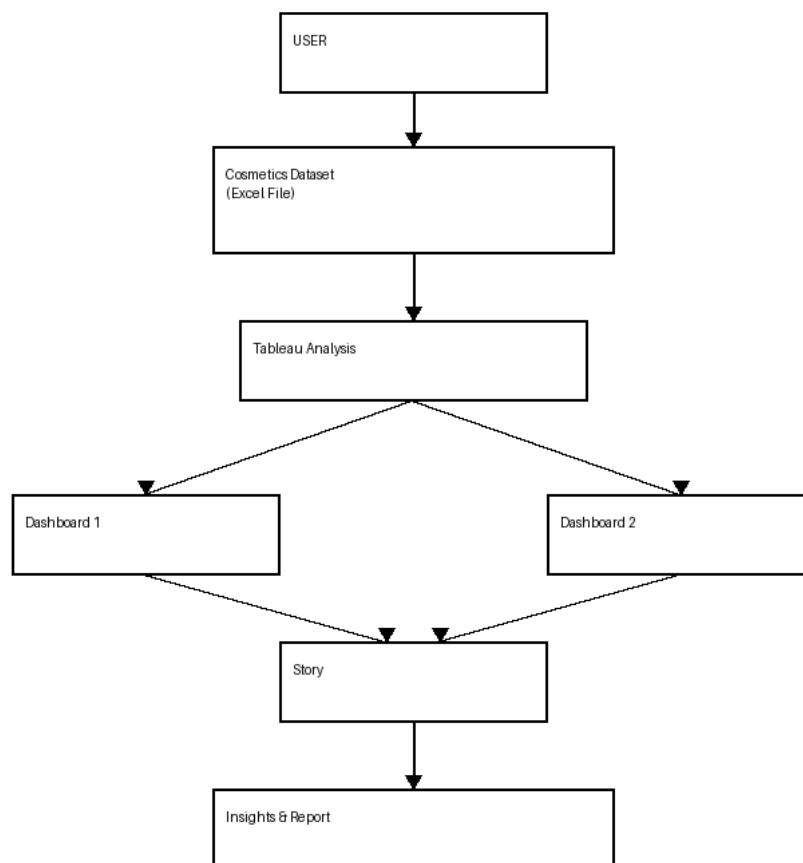
Secure handling of project data.

Availability and accessibility.

Scalability for future enhancements.

### 3.2 Data Flow Diagram

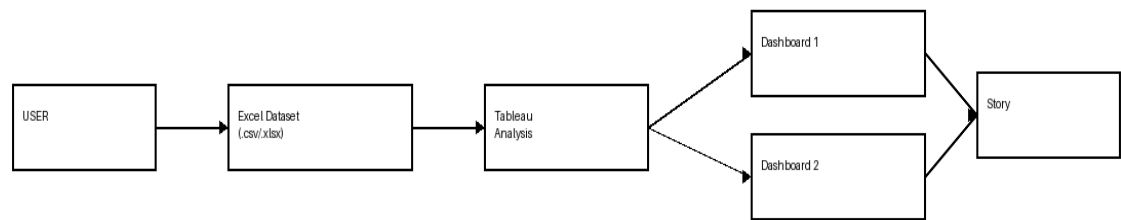
The Data Flow Diagram (DFD) represents the flow of cosmetic data through different stages of the system. The cosmetic dataset is collected from Excel files, processed and analyzed using Tableau, and then visualized through dashboards and stories. Finally, the insights are published through Tableau Public for user access and analysis.



### 3.3 Technology Stack

The project utilizes various technologies and tools to perform data collection, analysis, visualization, and publishing. Microsoft Excel is used for data preparation and preprocessing, while Tableau Public is used for data analysis, dashboard creation, story development, and publishing. The selected

technologies provide efficient and interactive data visualization capabilities for cosmetic market analysis.



| Component             | Technology Used |
|-----------------------|-----------------|
| Data Source           | Microsoft Excel |
| Data Analysis         | Tableau         |
| Visualization         | Tableau Public  |
| Dashboard Development | Tableau         |
| Story Creation        | Tableau         |
| Publishing Platform   | Tableau Public  |

## **4. PROJECT DESIGN**

The project design phase focuses on developing an effective solution for analyzing cosmetic products, brands, prices, and consumer preferences. During this phase, the project requirements, customer expectations, and business objectives were analyzed to design an interactive data visualization system using Tableau. The project design includes the proposed solution and the overall solution architecture, which define how the cosmetic data is collected, processed, analyzed, and visualized.

### **4.1 Proposed Solution**

The proposed solution for the "Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau" project is to develop an interactive analytics platform using Tableau. The system collects cosmetic product data from Excel datasets and performs data preprocessing, analysis, and visualization to generate meaningful insights.

The project analyzes various factors such as cosmetic brands, product categories, prices, skin types, and customer preferences. Interactive dashboards and stories are developed to help users understand market trends, compare products and brands, and identify customer behavior patterns. The proposed solution transforms raw data into interactive visualizations that support effective decision-making and business analysis.

The implementation of Tableau dashboards and stories enables users to explore cosmetic data efficiently and obtain valuable insights regarding market performance and consumer preferences.

### **4.2 Solution Architecture**

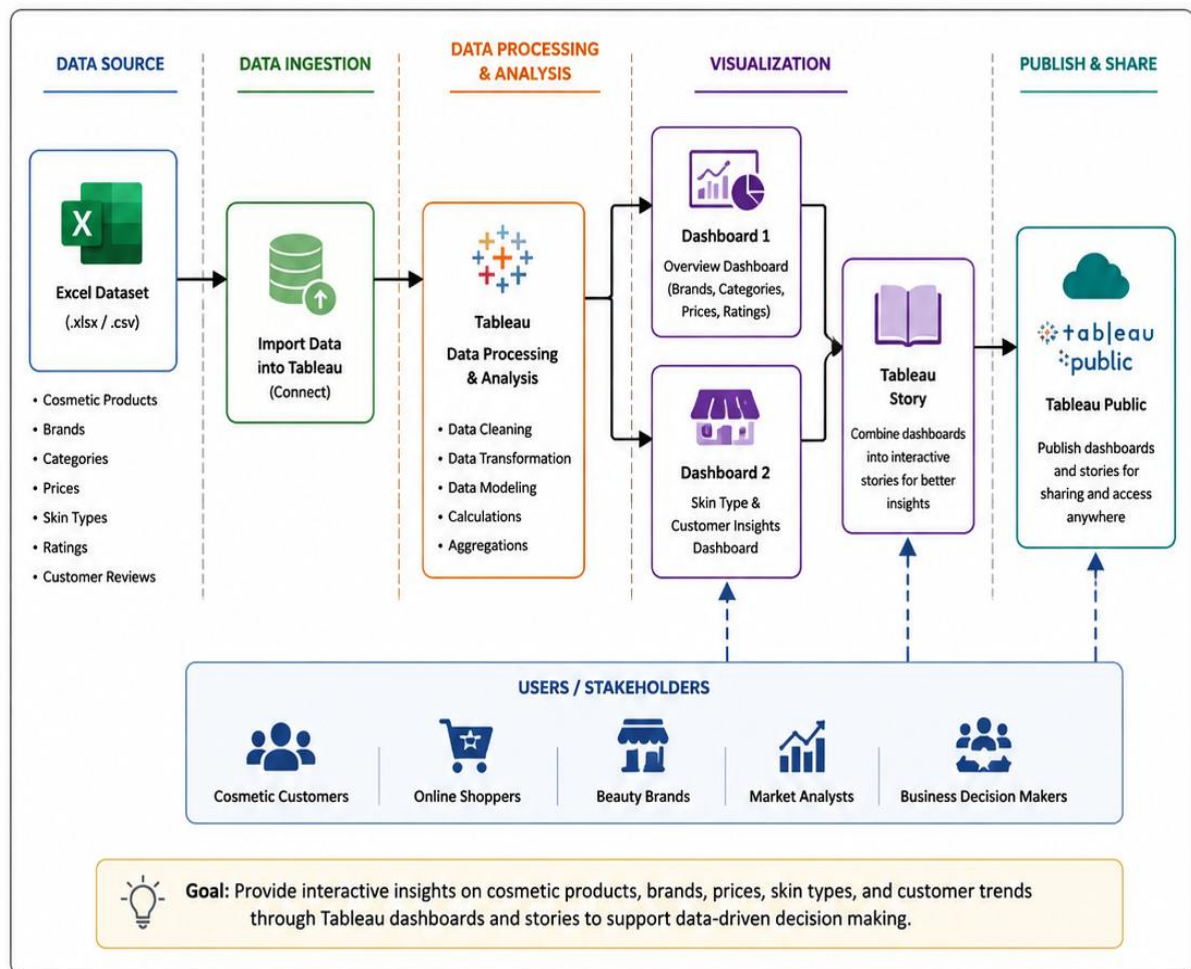
The solution architecture defines the structure and workflow of the Cosmetic Insights project. The system begins by collecting cosmetic product data from Excel datasets. The collected data is then imported into Tableau, where preprocessing, analysis, and visualization are performed.

Various charts, dashboards, and stories are created to analyze cosmetic brands, product categories, prices, skin types, and customer trends. The generated dashboards provide interactive visualizations that help users understand

market behavior and customer preferences. Finally, the dashboards and stories are published using Tableau Public for easy access and sharing.

The solution architecture provides a clear understanding of the interaction between different components and the flow of data throughout the project lifecycle.

**Solution Architecture Diagram:**



## 5. PROJECT PLANNING & SCHEDULING

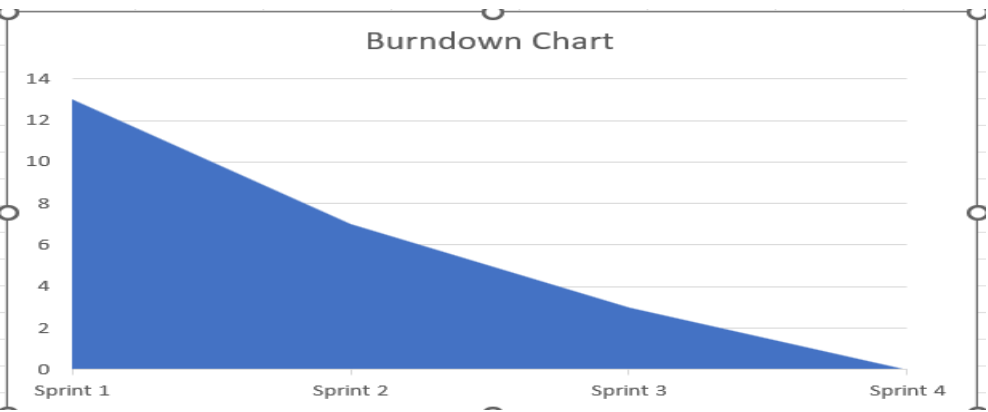
Project planning and scheduling play a significant role in the successful execution of a project. In the Cosmetic Insights project, the development process was divided into multiple phases to ensure systematic implementation and effective project management. The project activities were planned based on data collection, data analysis, dashboard development, story creation, publishing, and documentation.

The project was developed using a sprint-based approach, where each sprint focused on completing specific tasks within a defined period. This approach helped in monitoring project progress, managing resources efficiently, and ensuring timely completion of all project activities.

The project planning process included task allocation among team members, estimation of story points, sprint planning, progress tracking, and performance evaluation. The sprint-based methodology improved team collaboration and ensured effective completion of project objectives.

### Sprint Details

| Sprint   | Activities                                       |
|----------|--------------------------------------------------|
| Sprint 1 | Data Collection, Data Cleaning, Data Preparation |
| Sprint 2 | Data Analysis and Visualization                  |
| Sprint 3 | Dashboard Development                            |
| Sprint 4 | Story Creation, Publishing, Documentation        |



## 6. FUNCTIONAL AND PERFORMANCE TESTING

Functional and Performance Testing were conducted to evaluate the accuracy, reliability, and performance of the Cosmetic Insights project. The testing process ensured that the cosmetic dataset was successfully imported, processed, analyzed, and visualized using Tableau. Different parameters such as data rendering, preprocessing, utilization of filters, calculated fields, and chart generation were tested to verify the effectiveness of the system.

The testing process confirmed that the visualizations generated accurate insights related to cosmetic brands, prices, product categories, and skin types. Interactive filters and calculated fields functioned correctly and provided meaningful analytical results. The performance evaluation demonstrated that the system efficiently transformed raw cosmetic data into useful visual insights.

### 6.1 Performance Testing

| S.No | Parameter               | Description                                                    | Status |
|------|-------------------------|----------------------------------------------------------------|--------|
| 1    | Data Rendered           | Cosmetic dataset successfully imported and rendered in Tableau | Pass   |
| 2    | Data Preprocessing      | Data cleaning and preprocessing completed successfully         | Pass   |
| 3    | Utilization of Filters  | Brand, Price, and Skin Type filters applied successfully       | Pass   |
| 4    | Calculation Fields Used | Calculated fields generated accurate results                   | Pass   |
| 5    | Chart Generation        | Visualizations and charts generated successfully               | Pass   |
| 6    | Analysis Performance    | Cosmetic data analysis completed successfully                  | Pass   |

### Data Rendering

The cosmetic dataset was successfully imported into Tableau and rendered without any errors. All records and fields were properly loaded and validated for further analysis.

File Data Window Help

cosmetics (cosmetics.csv) Filters 0 | Add

Connections Add

cosmetics.csv Microsoft Excel

Sheets cosmetics

New Union

New Table Extension

cosmetics 13 fields 1472 rows 100 rows

Name cosmetics

Description No description available.

Fields

| Type | Field Name  | Physical Table | Remote Field ... |
|------|-------------|----------------|------------------|
| Abc  | Label       | cosmetics      | Label            |
| Abc  | Brand       | cosmetics      | Brand            |
| Abc  | Name        | cosmetics      | Name             |
| #    | Price       | cosmetics      | Price            |
| #    | Rank        | cosmetics      | Rank             |
| Abc  | Ingredients | cosmetics      | Ingredients      |
| #    | Combination | cosmetics      | Combination      |

| Label       | Brand              | Name                             | Price | Rank    | Ingredients                       |
|-------------|--------------------|----------------------------------|-------|---------|-----------------------------------|
| Moisturizer | LA MER             | Crème de la Mer                  | 175   | 4.10000 | Algae (Seaweed) Extract, Min...   |
| Moisturizer | SK-II              | Facial Treatment Essence         | 179   | 4.10000 | Galactomyces Ferment Filtra...    |
| Moisturizer | DRUNK ELEPHANT     | Protini™ Polypeptide Cream       | 68    | 4.40000 | Water, Dicaprylyl Carbonate, ...  |
| Moisturizer | LA MER             | The Moisturizing Soft Cream      | 175   | 3.80000 | Algae (Seaweed) Extract, Cyc...   |
| Moisturizer | IT COSMETICS       | Your Skin But Better™ CC+™...    | 38    | 4.10000 | Water, Snail Secretion Filtrat... |
| Moisturizer | TATCHA             | The Water Cream                  | 68    | 4.20000 | Water, Saccharomyces/Cam...       |
| Moisturizer | DRUNK ELEPHANT     | Lala Retro™ Whipped Cream        | 60    | 4.20000 | Water, Glycerin, Caprylic/ Ca...  |
| Moisturizer | DRUNK ELEPHANT     | Virgin Marula Luxury Facial Oil  | 72    | 4.40000 | 100% Unrefined Sclerocraya ...    |
| Moisturizer | KIEHL'S SINCE 1851 | Ultra Facial Cream               | 29    | 4.40000 | Water, Glycerin, Cyclohexasil...  |
| Moisturizer | LA MER             | Little Miss Miracle Limited-E... | 325   | 5.00000 | Algae (Seaweed) Extract, Min...   |
| Moisturizer | FRESH              | Lotus Youth Preserve Moistur...  | 45    | 4.30000 | Water, Glycerin, Propylene GL...  |
| Moisturizer | KIEHL'S SINCE 1851 | Midnight Recovery Concentr...    | 47    | 4.40000 | Cantharidin/Cantharidin Trieth... |

Data Source price range All skin types Sheet 2 Sheet 3 Sheet 4 Sheet 5 sensitive Sheet 6 Sheet 7 Sheet 8 Sheet 9 Sheet 10 Sheet 11 Sheet 12 Sheet 13 Sheet 14 Sheet 15 Dashboard 1 Dashboard 2 Story 1 Sai Durga Amr...

## Data Preprocessing

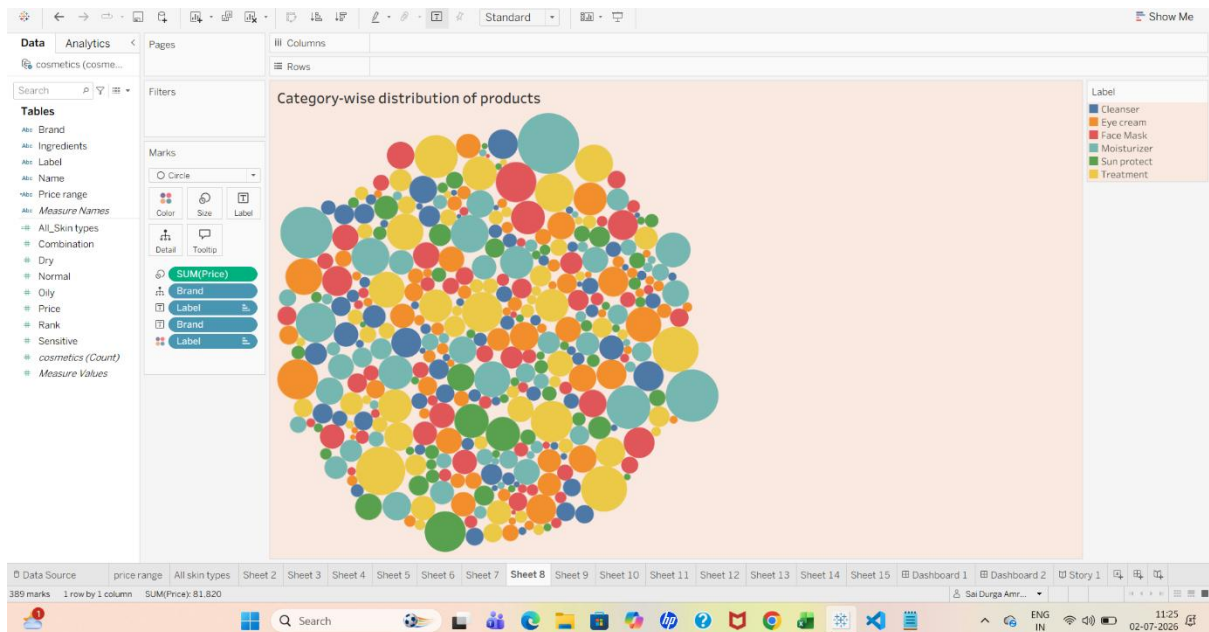
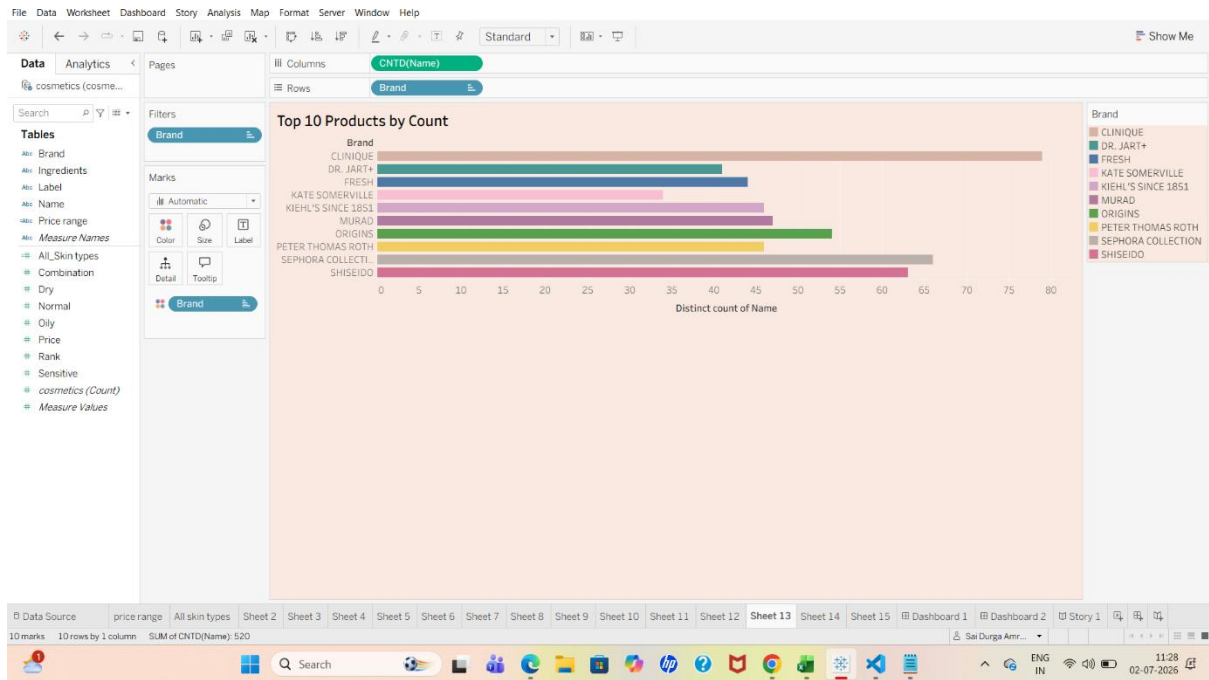
Data preprocessing techniques such as data cleaning, validation, and calculated field creation were performed successfully to improve data quality and analytical accuracy.

## Filter Utilization

Interactive filters including Brand Filter, Price Filter, and Skin Type Filter were implemented successfully to enable dynamic exploration of cosmetic data.

## Chart Generation

Various charts and visualizations were created successfully to analyze cosmetic brands, product categories, pricing trends, and customer preferences.



## Performance Evaluation

The testing results indicate that the Cosmetic Insights project successfully achieved its objectives. The visualizations generated accurate insights and provided effective analytical support for understanding cosmetic market trends and consumer behavior.

# 7. OUTPUT RESULTS

The Cosmetic Insights project successfully generated various visualizations, dashboards, and stories using Tableau. The output results provide meaningful insights into cosmetic products, brands, pricing patterns, skin types, and consumer preferences. The developed visualizations helped analyze market trends and supported effective decision-making through interactive analytics.

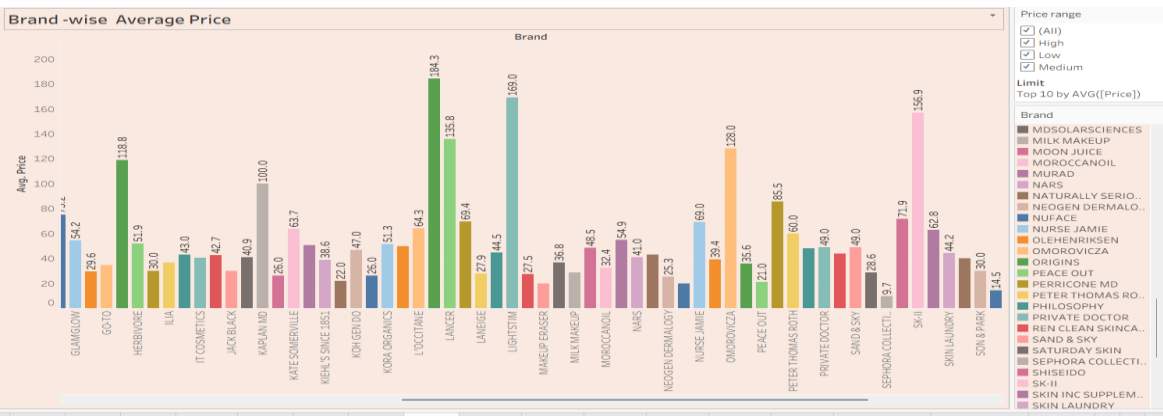
The project successfully transformed raw cosmetic data into interactive visual representations, enabling effective analysis of cosmetic products and customer behavior. The generated dashboards and stories provided comprehensive insights into cosmetic market trends and consumer preferences.

## 7.1 Chart Visualizations

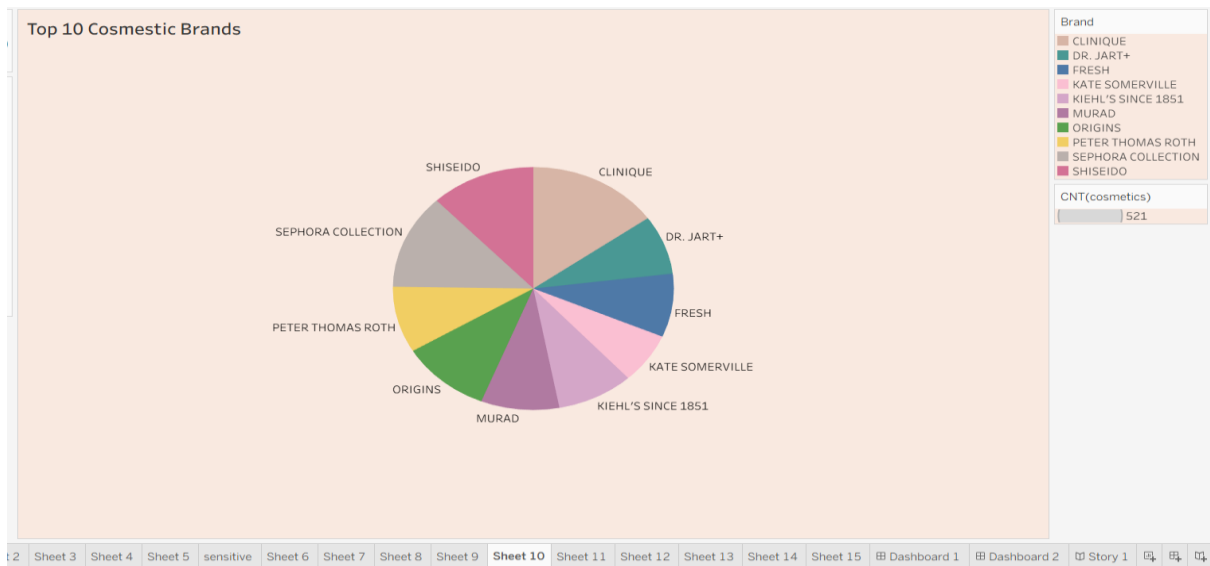
The project includes multiple chart visualizations that analyze different aspects of cosmetic products, brands, prices, and customer preferences.



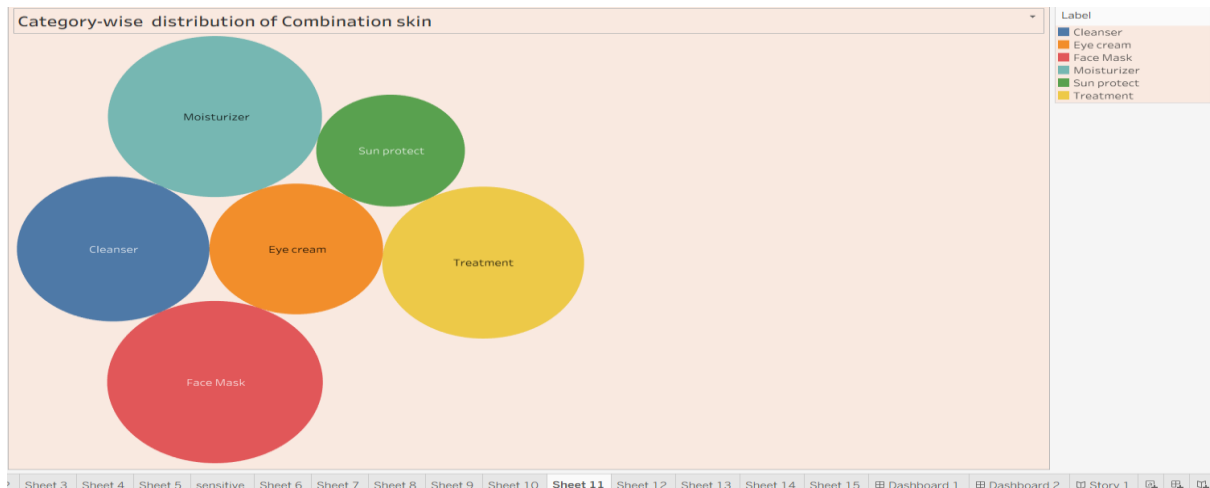
Category-wise Distribution of Cosmetic Products



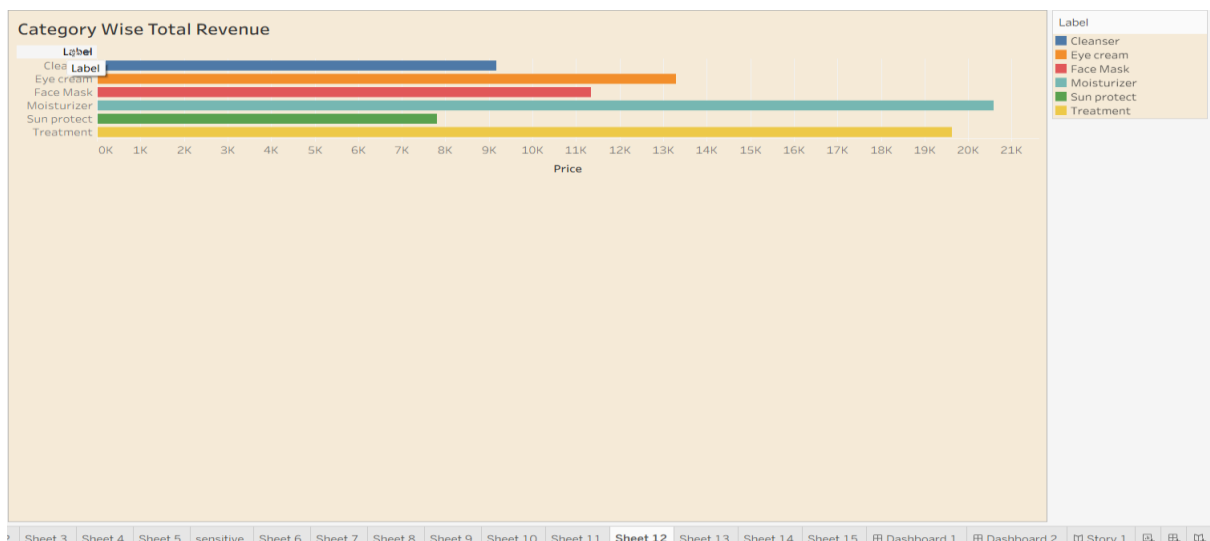
Brand-wise Average Price Analysis



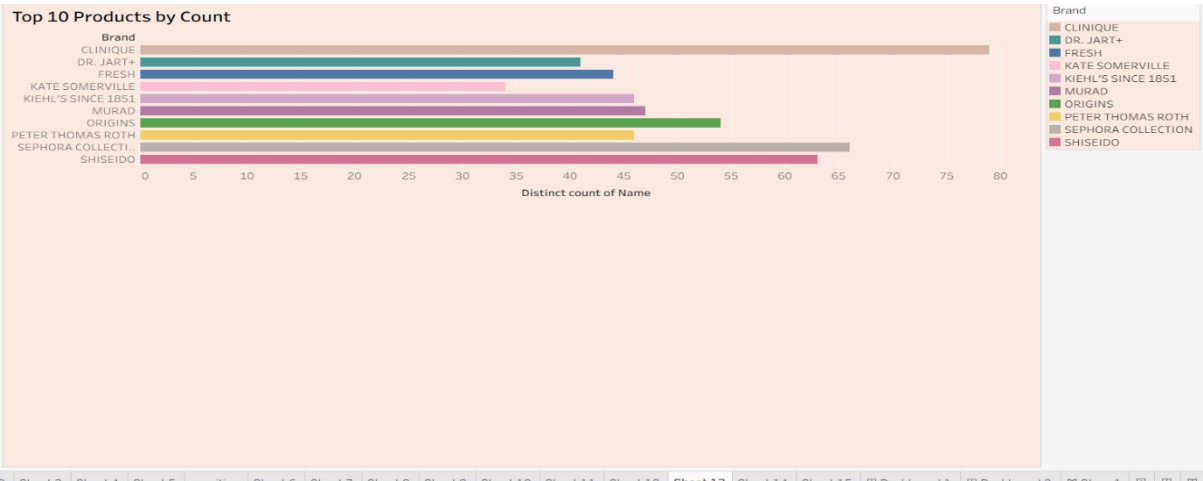
Top 10 Cosmetic Brands Analysis



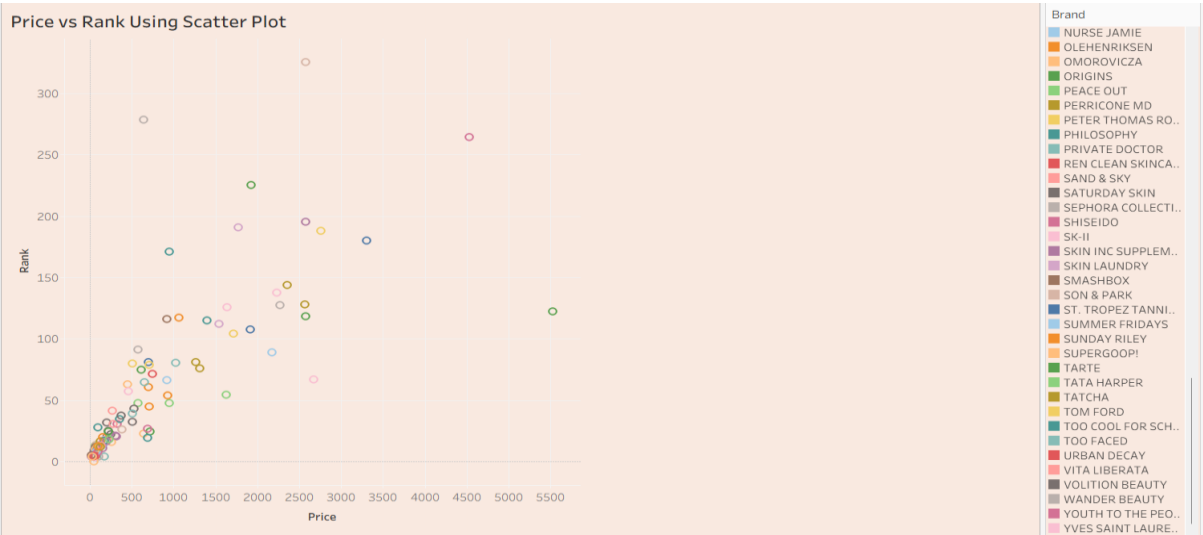
Category-wise Distribution of Combination Skin Products



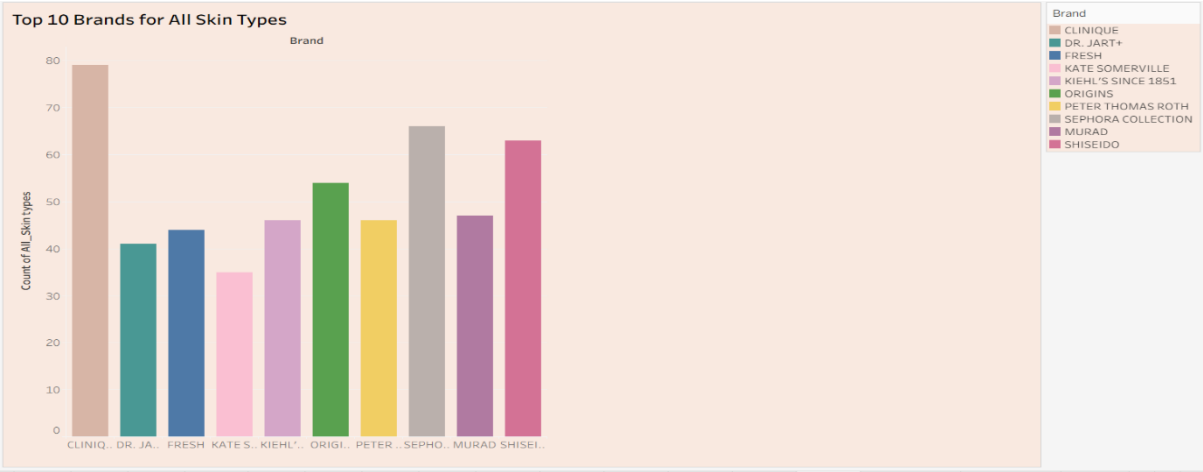
Top 10 Brands Suitable for All Skin Types



Category-wise Total Revenue Analysis



Price vs Rank Analysis using Scatter Plot



Top 10 Products by Product Count

## 7.2 Dashboard Screenshots

The dashboards developed in the Cosmetic Insights project provide interactive and comprehensive visual analysis of cosmetic products, brands, prices, skin types, and consumer preferences. These dashboards help users understand market trends, compare products, analyze customer requirements, and make effective business decisions. The dashboards transform raw cosmetic data into meaningful visual insights through interactive charts, filters, and analytical tools..

### **Dashboard 1: Cosmetic Insights Dashboard 1**

Dashboard 1 provides a comprehensive overview of cosmetic products and consumer insights through interactive visualizations and analytical charts. The dashboard includes various visualizations related to category-wise product distribution, brand-wise average price analysis, top cosmetic brands, and combination skin product analysis. These visualizations help users understand the distribution and performance of cosmetic products across different categories and brands.

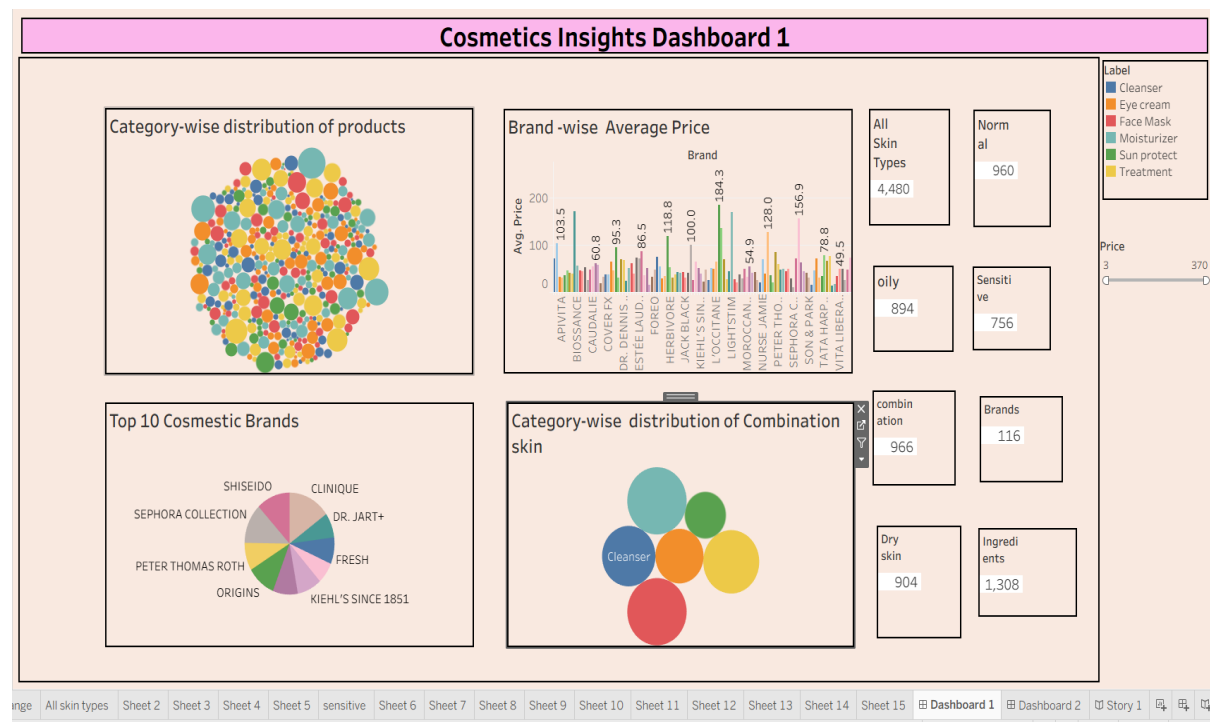
The dashboard enables users to analyze cosmetic products based on categories, brands, prices, and skin types through interactive filters and visualization tools. The category-wise analysis helps identify the most popular cosmetic categories, while the brand-wise average price analysis provides insights into pricing strategies adopted by different cosmetic brands. The top cosmetic brands analysis highlights the leading brands based on product performance and market presence.

The combination skin product analysis provides detailed information about products suitable for combination skin types, helping users understand consumer preferences and product suitability. The interactive filters allow users to explore cosmetic data dynamically and generate customized insights based on specific requirements.

Overall, Dashboard 1 provides meaningful insights into cosmetic market trends, product performance, pricing strategies, and consumer behavior. The dashboard supports effective decision-making and enhances the understanding of cosmetic industry trends through interactive and comprehensive data visualization.

## Tableau Public Link :

[https://public.tableau.com/views/CosmeticInsightsDashboard1/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticInsightsDashboard1/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)



## Dashboard 2: Cosmetic Insights Dashboard 2

Dashboard 2 provides detailed analysis of cosmetic products, consumer preferences, and market trends through interactive visualizations. The dashboard includes various analytical charts such as top brands suitable for all skin types, category-wise revenue analysis, top products by product count, and price versus rank analysis using scatter plots. These visualizations help users understand the performance of cosmetic products across different categories and identify important market trends.

The dashboard enables comprehensive analysis of product popularity, pricing patterns, customer requirements, and brand performance. It provides insights into how different cosmetic brands perform in the market and how customer preferences vary based on product categories and skin types. The revenue analysis helps identify high-performing categories, while the product count analysis highlights the most frequently available cosmetic products.

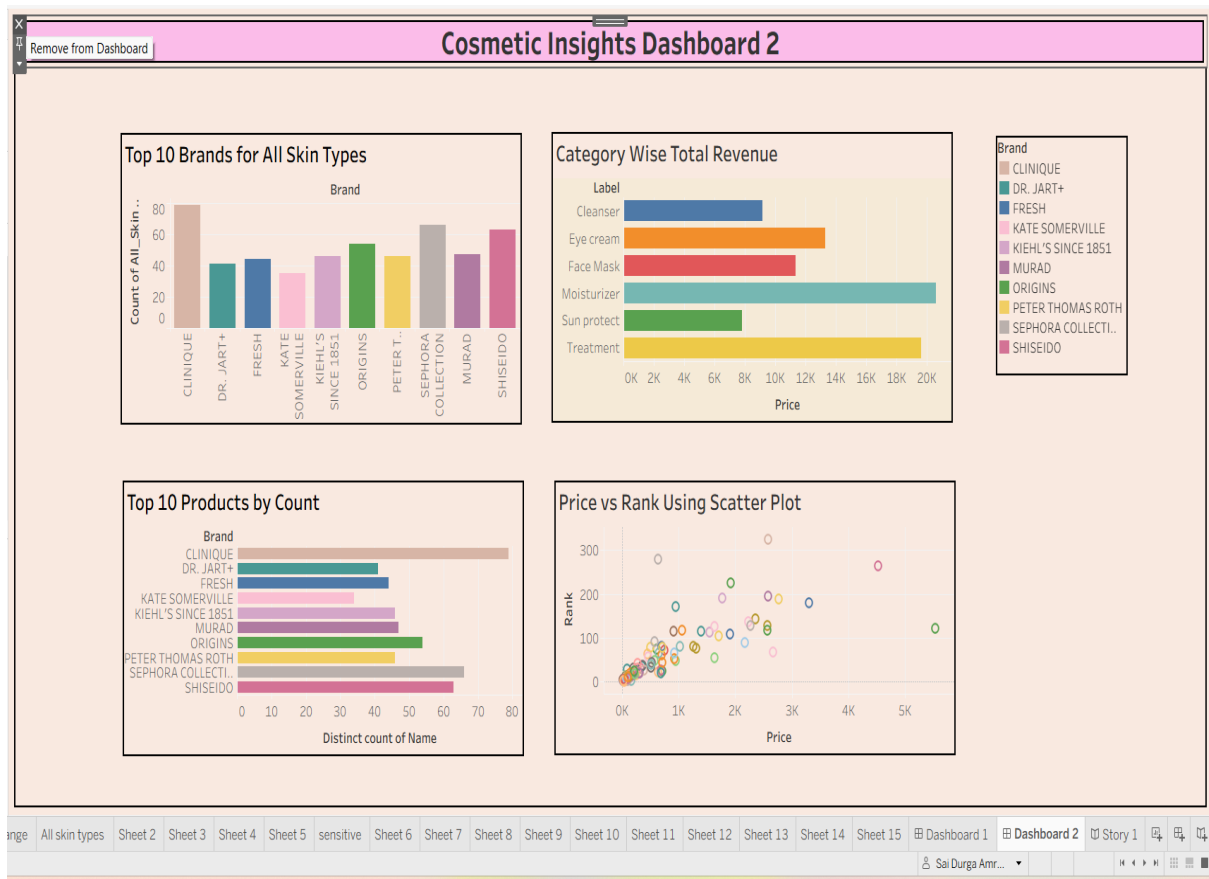
The scatter plot visualization assists in understanding the relationship between product price and product ranking, allowing users to evaluate pricing strategies and market competitiveness. Interactive filters further enhance the user

experience by enabling dynamic exploration of cosmetic data based on various parameters.

Overall, Dashboard 2 provides meaningful insights into cosmetic market behavior, customer preferences, product performance, and business opportunities. The dashboard supports effective decision-making and helps users understand current trends in the cosmetic industry through interactive and comprehensive data visualization.

### Tableau Public Link:

[https://public.tableau.com/views/CosmeticInsightsDashboard2/Dashboard2?:language=enUS&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticInsightsDashboard2/Dashboard2?:language=enUS&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)



## 7.3 Story Screenshot

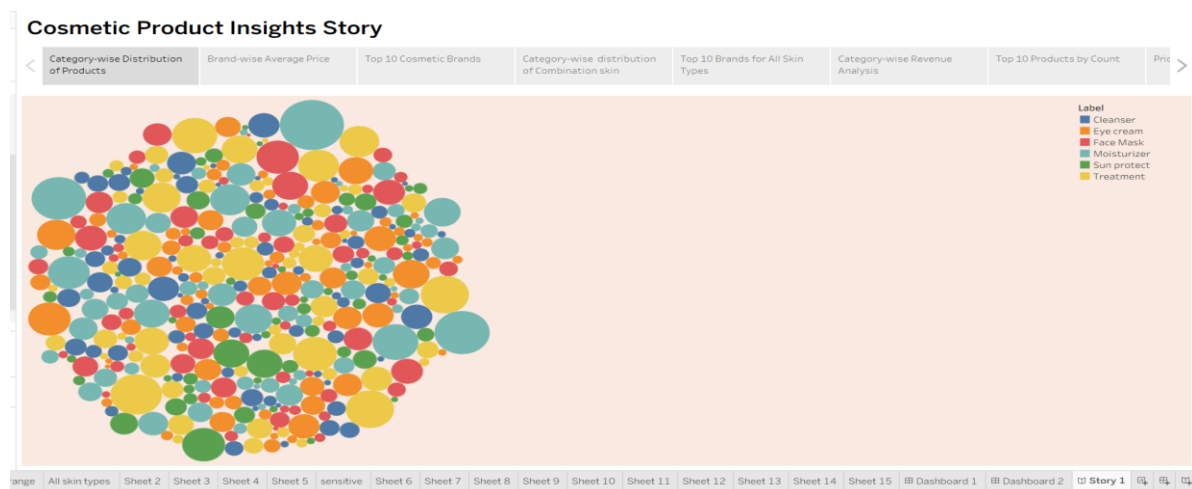
The Tableau Story integrates multiple dashboards and visualizations into a structured and interactive presentation format. It provides a comprehensive analysis of cosmetic products, market trends, pricing patterns, customer preferences, skin type analysis, and brand performance. The story presents the complete flow of data analysis in a systematic and sequential manner, enabling users to understand the relationships between various cosmetic factors and consumer behavior.

The story combines different dashboards and visualizations to provide meaningful insights into cosmetic products and market performance. It helps users analyze product categories, compare brand performance, evaluate pricing strategies, and understand customer preferences based on different skin types. The integrated visualizations support effective interpretation of analytical results and provide a clear understanding of cosmetic market trends.

The Tableau Story also enables users to explore data interactively and identify important business opportunities through visual analytics. By presenting multiple dashboards and charts in a logical sequence, the story improves data understanding and supports effective decision-making. The story visualization provides comprehensive insights into the cosmetic industry, consumer behavior, product performance, and market trends through interactive and user-friendly visualizations.

### Tableau Public Link:

[https://public.tableau.com/views/CosmeticProductInsightsStory/Story1?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticProductInsightsStory/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)



## 7.4 Tableau Public Profile

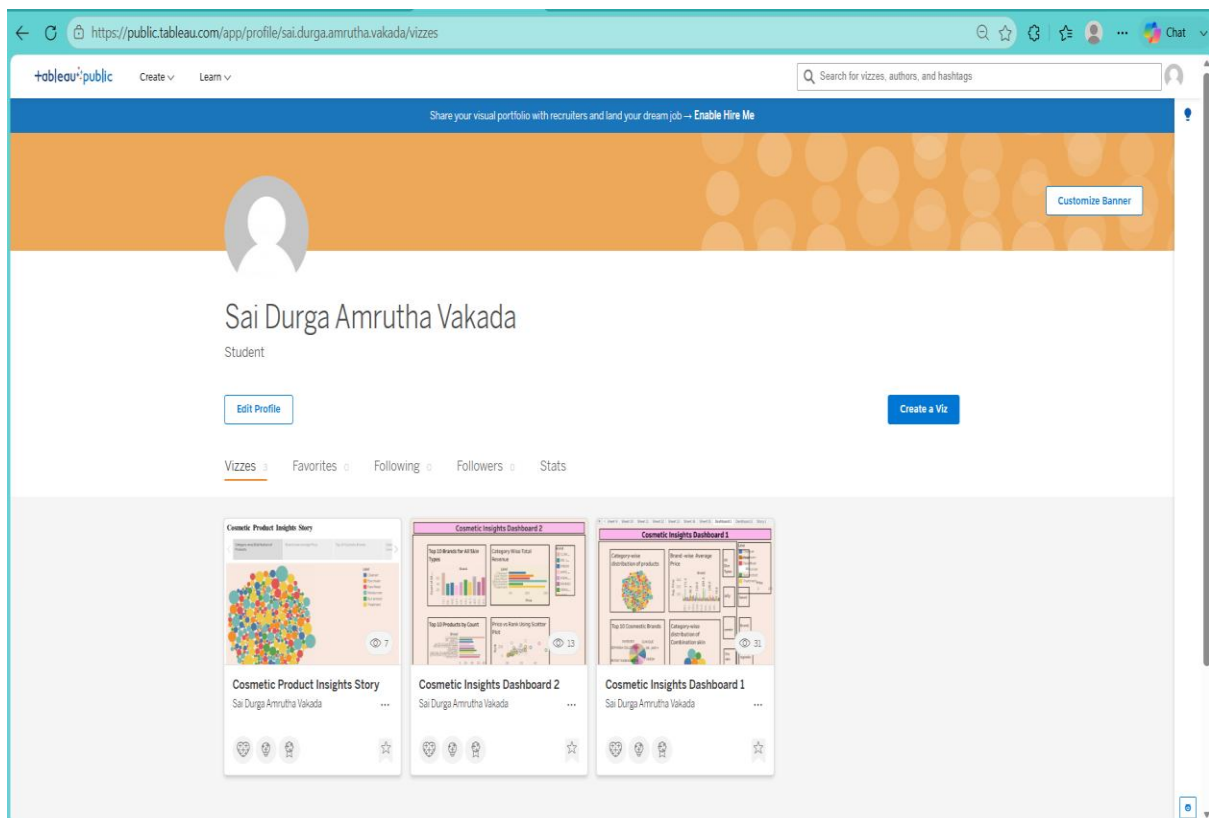
The Tableau Public profile serves as an online platform for publishing and accessing the dashboards and story visualizations developed in the Cosmetic Insights project. The profile contains all the published visualizations created during the project implementation process and provides interactive access to the project outputs.

The Tableau Public profile enables users to explore cosmetic product analysis, market trends, pricing patterns, customer preferences, and skin type analysis through interactive visualizations. It also supports online sharing, accessibility, and effective presentation of the developed dashboards and story.

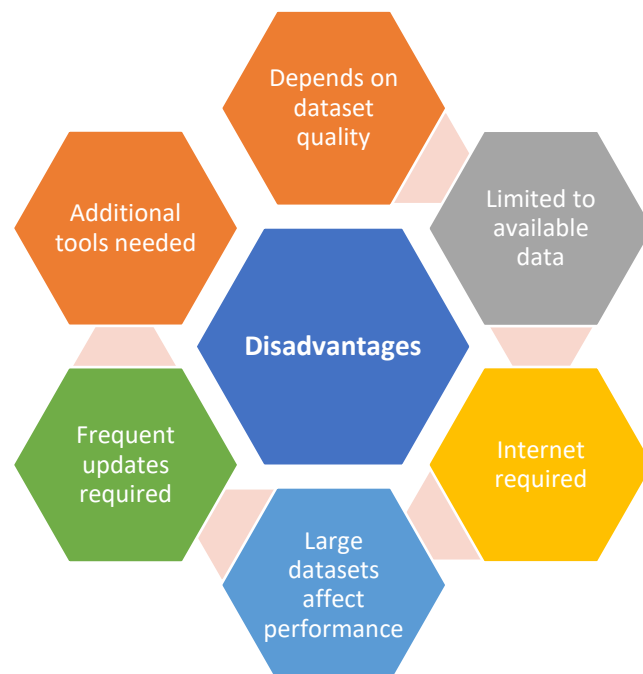
Overall, the Tableau Public profile acts as a centralized repository for the visual analytics developed in the Cosmetic Insights project and enhances the accessibility and usability of the project outputs.

### Tableau Public Profile Link:

<https://public.tableau.com/app/profile/sai.durga.amrutha.vakada/vizzes>



## 8. ADVANTAGES & DISADVANTAGES



## 9. CONCLUSION

The Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau project was successfully completed using Tableau data visualization techniques. The project effectively analyzed cosmetic products, brands, pricing patterns, skin types, and customer preferences through interactive charts, dashboards, and story visualizations.

The developed visualizations provided meaningful insights into cosmetic market trends and consumer behavior. The interactive dashboards and story helped in understanding product performance, brand analysis, pricing strategies, and customer requirements in an effective and user-friendly manner.

The project successfully transformed raw cosmetic data into interactive visual analytics, enabling better understanding and decision-making. Overall, the Cosmetic Insights project demonstrated the effectiveness of Tableau in data analysis, visualization, and business intelligence applications and achieved the intended project objectives successfully.

## 10. FUTURE SCOPE

The Cosmetic Insights project can be further enhanced by incorporating advanced analytical techniques and additional data sources. Future improvements can help provide more accurate insights, better user experience, and enhanced decision-making capabilities.

The project can be extended by integrating real-time cosmetic market data, customer reviews, and product recommendation systems. Advanced technologies such as Machine Learning and Artificial Intelligence can be implemented to predict customer preferences, product trends, and market behavior.

Additional features such as personalized product recommendations, sentiment analysis of customer reviews, and predictive analytics can further improve the effectiveness of the system. The dashboards and visualizations can also be enhanced by including more interactive features and larger datasets.

Future enhancements may also include mobile accessibility, cloud-based analytics, and integration with e-commerce platforms to provide comprehensive insights into the cosmetic industry. Overall, the project has significant potential for future development and can support advanced business intelligence and consumer analytics applications.

# 11. APPENDIX

## Dataset Link:

<C:\Users\vakada amrutha\OneDrive\Desktop\data\cosmetics.csv.xlsx>

## Dashboard 1 Tableau Public Link:

[https://public.tableau.com/views/CosmeticInsightsDashboard1/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticInsightsDashboard1/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

## Dashboard 2 Tableau Public Link:

[https://public.tableau.com/views/CosmeticInsightsDashboard2/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticInsightsDashboard2/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

## Tableau Story Link:

[https://public.tableau.com/views/CosmeticProductInsightsStory/Story1?:language=en-US&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/CosmeticProductInsightsStory/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

## Tableau Public Profile Link:

<https://public.tableau.com/app/profile/sai.durga.amrutha.vakada/vizzes>

## Project Demo Video Link:

[https://drive.google.com/file/d/1sTFd8xRhWuZmc\\_UJ0PrfJVount-lc-4y/view?usp=drive\\_link](https://drive.google.com/file/d/1sTFd8xRhWuZmc_UJ0PrfJVount-lc-4y/view?usp=drive_link)

**-----THANK YOU-----**